MSG No. 196.4 $F23$ [Type I, cubic]

Table 1: Wyckoff site: `4a`, site symmetry: `23.`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `4b`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24]`
3	$\left[0, \frac{1}{2}, 0\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36]`
4	$\left[0, 0, \frac{1}{2}\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48]`

Table 3: Wyckoff site: `4c`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,5,6,14,22,23,27,31,36,40,44,45]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[2,10,11,13,17,18,28,32,33,39,43,48]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,7,12,16,20,21,25,29,30,38,46,47]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[4,8,9,15,19,24,26,34,35,37,41,42]`

Table 4: Wyckoff site: `4d`, site symmetry: `23.`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[1,5,6,14,22,23,27,31,36,40,44,45]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[2,10,11,13,17,18,28,32,33,39,43,48]`
3	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[3,7,12,16,20,21,25,29,30,38,46,47]`
4	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[4,8,9,15,19,24,26,34,35,37,41,42]`

Table 5: Wyckoff site: `16e`, site symmetry: `.3.`

No.	position	mapping
1	$[x, x, x]$	`[1,5,6]`
2	$[x, -x, -x]$	`[2,10,11]`
3	$[-x, x, -x]$	`[3,7,12]`
4	$[-x, -x, x]$	`[4,8,9]`
5	$\left[x, x+\frac{1}{2}, x+\frac{1}{2}\right]$	`[13,17,18]`
6	$\left[x, \frac{1}{2}-x, \frac{1}{2}-x\right]$	`[14,22,23]`
7	$\left[-x, x+\frac{1}{2}, \frac{1}{2}-x\right]$	`[15,19,24]`
8	$\left[-x, \frac{1}{2}-x, x+\frac{1}{2}\right]$	`[16,20,21]`

continued ...

Table 5

No.	position	mapping
9	$\left[x+\frac{1}{2},\, x,\, x+\frac{1}{2}\right]$	[25,29,30]
10	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-x\right]$	[26,34,35]
11	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}-x\right]$	[27,31,36]
12	$\left[\frac{1}{2}-x,\, -x,\, x+\frac{1}{2}\right]$	[28,32,33]
13	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x\right]$	[37,41,42]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -x\right]$	[38,46,47]
15	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, -x\right]$	[39,43,48]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, x\right]$	[40,44,45]

Table 6: Wyckoff site: 24f, site symmetry: 2..

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,2]
2	$[-x,\, 0,\, 0]$	[3,4]
3	$[0,\, x,\, 0]$	[5,12]
4	$[0,\, 0,\, x]$	[6,9]
5	$[0,\, 0,\, -x]$	[7,11]
6	$[0,\, -x,\, 0]$	[8,10]
7	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[13,14]
8	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[15,16]
9	$\left[0,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[17,24]
10	$\left[0,\, \frac{1}{2},\, x+\frac{1}{2}\right]$	[18,21]
11	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-x\right]$	[19,23]
12	$\left[0,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[20,22]
13	$\left[x+\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	[25,26]
14	$\left[\frac{1}{2}-x,\, 0,\, \frac{1}{2}\right]$	[27,28]
15	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[29,36]
16	$\left[\frac{1}{2},\, 0,\, x+\frac{1}{2}\right]$	[30,33]
17	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-x\right]$	[31,35]
18	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[32,34]
19	$\left[x+\frac{1}{2},\, \frac{1}{2},\, 0\right]$	[37,38]
20	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, 0\right]$	[39,40]
21	$\left[\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[41,48]
22	$\left[\frac{1}{2},\, \frac{1}{2},\, x\right]$	[42,45]
23	$\left[\frac{1}{2},\, \frac{1}{2},\, -x\right]$	[43,47]
24	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[44,46]

Table 7: Wyckoff site: 24g, site symmetry: 2..

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	[1,14]
2	$\left[x,\, \frac{3}{4},\, \frac{3}{4}\right]$	[2,13]
3	$\left[-x,\, \frac{1}{4},\, \frac{3}{4}\right]$	[3,16]
4	$\left[-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[4,15]

continued ...

Table 7

No.	position	mapping
5	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,36]
6	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[6,45]
7	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[7,47]
8	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[8,34]
9	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	[9,42]
10	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[10,32]
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[11,43]
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[12,29]
13	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[17,48]
14	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[18,33]
15	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[19,35]
16	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[20,46]
17	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[21,30]
18	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[22,44]
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[23,31]
20	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[24,41]
21	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[25,38]
22	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[26,37]
23	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[27,40]
24	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[28,39]

Table 8: Wyckoff site: 48h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[x, -y, -z]$	[2]
3	$[-x, y, -z]$	[3]
4	$[-x, -y, z]$	[4]
5	$[z, x, y]$	[5]
6	$[y, z, x]$	[6]
7	$[-y, z, -x]$	[7]
8	$[-z, -x, y]$	[8]
9	$[-y, -z, x]$	[9]
10	$[z, -x, -y]$	[10]
11	$[y, -z, -x]$	[11]
12	$[-z, x, -y]$	[12]
13	$\left[x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[13]
14	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[14]
15	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[15]
16	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[16]
17	$\left[z, x+\frac{1}{2}, y+\frac{1}{2}\right]$	[17]
18	$\left[y, z+\frac{1}{2}, x+\frac{1}{2}\right]$	[18]
19	$\left[-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$	[19]
20	$\left[-z, \frac{1}{2}-x, y+\frac{1}{2}\right]$	[20]
21	$\left[-y, \frac{1}{2}-z, x+\frac{1}{2}\right]$	[21]

continued ...

Table 8

No.	position	mapping
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[22]
23	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23]
24	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[24]
25	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[27]
28	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[28]
29	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[29]
30	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[30]
31	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[31]
32	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[32]
33	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[33]
34	$\left[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y\right]$	[34]
35	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x\right]$	[35]
36	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[36]
37	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[37]
38	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[38]
39	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[39]
40	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[40]
41	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[41]
42	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[42]
43	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x\right]$	[43]
44	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[44]
45	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[45]
46	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[46]
47	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[47]
48	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y\right]$	[48]